Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/12/2006
PI Name:	Lee, Jerry Ssu-Hsien Ph.D.		
Project Title:	Investigating the effects of polarity on protein expression and post-translational modification in cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Cell polarity mediates intracellular organelle arrangement through the microtubule-organizing center (MTOC). Loss of microtubule structure and the MTOC has been shown to occur in cells during space launch, and although both return during spaceflight, the cells exhibit growth arrest and increased apoptosis. MTOC repositioning causes reorientation of the Golgi towards the anterior of the cell, which may result in changes in protein expression and post- translation modifications. Using a novel mechanical shear assay, I have demonstrated that Swiss 3T3 fibroblasts polarize in the direction of flow in a Cdc42 dependent manner. Thus, I propose to study the effects of cell polarity, and subsequent organelle reposition, on protein expression and modification in cells. This would provide insight into understanding, and overcoming, the short and long term effects of space flight on humans.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	To provide insight into understanding, and overcoming, the short and long term effects of space flight on humans, I have chosen to study these effects on a cellular basis using shear flow to mimic effects of space flight. I have previously documented my discovery of changes in cell polarity. A natural progression included the study of changes in cellular mechanics due to fluid flow. As outlined last year, I have successfully modified our lab's technique of intracellular microrheology (Kole et al. 2004, Mol. Bio. Cell) through incorporation of ballistic introduction of fluorescent particles. The process, termed ballistic intracellular nanorheology, or BIN for short, was used to monitor changes in cellular mechanics. Cells in vivo are constantly subjected to mechanical stresses that play important regulatory roles in various physiological and pathological processes. These forces are compounded during space launch and may be diminished or absent in microgravity space flight. Cytoskeletal reorganizations that occur in response to forces have been studied extensively, but whether the cytoplasm of an adherent cell adapts its mechanical properties to respond to fluid stress is largely unknown. To assess these changes, we use fluid flow to stimulate cells and apply BIN to probe, with high resolution, possible local viscoelastic changes in the cytoplasm of individual cells subjected to fluid flow. Our assasy reveals that shear flow induces a dramatic sustained 25-fold increase in cytoplasmic viscosity in serum-starved Swiss 3T3 fibroblasts. By contrast, cells stimulated with the actin contractile agonist, LPA, show highly transient stiffening of much lower amplitude, despite the formation of similar cytoskeletal structures. Shear-induced cytoplasmic stiffening is attenuated by inhibiting actomyosin interactions and is entirely eliminated by specific Rho-kinase (ROCK) inhibition. Together, these results show that biochemical and biophysical stimuli may elicit the formation of qualitatively similar cytoskeleton structures (i.e. stress fibers and focal adhesions), but induces quantitatively different micromechanical responses. Our results suggest that when an adherent cell is subjected to shear stresses, its first order of action is to prevent detachment from its substratum by greatly stiffening its cytoplasm through enhanced actin assembly and Rho-kinase mediated contractility. These results were submitted to the peer-reviewed journal, Journal of Cell Science, and were accepted for publication in May 2006. I then combined both of my previous works in cell polarity and cell mechanics by studying effects of nuclear lamin a/c deficiency. Rather than working with transiently transfected cells (Lee et al. 2005, Mol. Bio. Cell), I now had both wild-type and -/- knockout mouse embryo fibroblasts (MEFs) to work with. We found that knocking out lamin a/c resulted in cells unable to polarize and also significantly weaker in mechanics. The results are currently being reviewed and further studies using fluid flow are being conducted by other students.
Bibliography Type:	Description: (Last Updated: 09/01/2009)
Abstracts for Journals and Proceedings	Lee JS, Panorchan P, Tseng Y, Wirtz D. "Intracellular Response to Shear Flow is Mediated by Rho-kinase." American Society for Cell Biology Annual Meeting, San Francisco, December 2005. Handout Booklet for American Society for Cell Biology Annual Meeting, 2005. , Dec-2005
Articles in Peer-reviewed Journals	Tseng Y, Kole TP, Lee JS, Fedorov E, Almo SC, Schafer BW, Wirtz D. "How actin crosslinking and bundling proteins cooperate to generate an enhanced cell mechanical response." Biochem Biophys Res Commun. 2005 Aug 19;334(1):183-92. PMID: 15992772 , Aug-2005
Articles in Peer-reviewed Journals	Panorchan P, Lee JS, Kole TP, Tseng Y, Wirtz D. "Microrheology and ROCK signaling of human endothelial cells embedded in a 3D matrix." Biophys J. 2006 Nov 1;91(9):3499-507. PubMed PMID: 16891369 , Nov-2006
Articles in Peer-reviewed Journals	Lee JS, Panorchan P, Hale CM, Khatau SB, Kole TP, Tseng Y, Wirtz D. "Ballistic intracellular nanorheology reveals ROCK-hard cytoplasmic stiffening response to fluid flow." J Cell Sci. 2006 May 1;119(Pt 9):1760-8. PMID: 16636071 , May-2006